

THE GRADIENT SCORE

Independent Engineering Audit

Does NowTrendIn's data accurately reflect the current market and its "human attention, before it arrives" claim?

Subject	NowTrendIn 2.0 Gradient Score, the Detection/Confidence gap, and Market Signal tiers
Reviewer frame	Engineer auditing for a venture-capital / investment-bank diligence team
Method	24 AI agents: intended-vs-actual model mapping (code = source of truth), live-data diagnosis, adversarially-verified no-gap falsification test, external-benchmark research
Data basis	Live engine snapshot 2026-06-22/23: 2,103 scored topics, a 100-topic stratified no-gap sample, 88 Market Signal instruments. Single-day - directionally decisive, not a multi-week out-of-sample validation.
Operating period	~2-4 weeks of live scoring at time of audit
Date	2026-06-23

Integrity stance of this audit. Findings are reported to the product's own standard: accuracy over impressiveness, no circular metrics, reputable/licensed sources only, and measurement - not advice. Where the engine is honest and correct, this report says so; where it fails, it cites the live numbers and the basis for the determination.

Executive Summary

VERDICT: NOT YET DEFENSIBLE AS AN UNQUALIFIED "ATTENTION BEFORE IT ARRIVES" INSTRUMENT.

The instrument fails its own no-gap credibility floor on the day's single most important confirmed event and on a systematic class of already-arrived topics - while the core "lead" claim remains unproven (accuracy ledger at 0.0% honest hit rate). The design is structurally honest and the top tier works; the mid-mainstream band and the market tiers do not.

Bottom line: NowTrendIn's instrument partially reflects human attention today, but it FAILS the no-gap test on the data's most important live data point and on a systematic class of already-arrived topics. On 2026-06-22 Alan Greenspan died at 100 - the day's premier, globally wall-to-wall mainstream event - and the engine scored it Detection 91.8 / Confidence 75.6 / gap +16.2, the single largest positive gap in the 100-topic sample, landing a confirmed obituary in the "very early, not-yet-confirmed" Anomalies band.

That is precisely the broken behavior the product's own no-gap principle says invalidates the gap metric. The failure is systemic, not a fluke: every mainstream nation and mega-cap outside a privileged top tier collapses to a near-identical ~45 Detection / ~67 Confidence signature badged EMERGING (china, india, russia, france, apple, openai, korea), with a uniform ~22 LAGGING gap - Confidence systematically exceeding Detection on topics that have unambiguously already arrived, the inverse of the intended "early = Detection leads" design.

The single biggest risk is therefore credibility at the already-arrived end of the curve: the dual-pathway anti-collapse gate demonstrably works at the very top (trump 94.6/94.2, FIFA 93.7/93.7, iran 86.1/90.1, election 94.7/94.3) but does not extend across the full mainstream band. Separately, a real data-integrity bug exists: the stored heisenberg_gap field contradicts its own Detection-Confidence on the entire mcp family (det-conf ~ +1.5 but stored gap -23), directly breaking the "both scores explain each other" test.

The good news that must be weighed against this: the scores DO discriminate (monotonic stage ladder, 520/595 distinct values, gap converging to ~0 only for genuinely-arrived BREAKOUT topics), the engine does NOT fabricate confidence for niche topics (a one-directional failure mode), and the no-N-into-Gradient firewall - the core integrity rule - is correctly enforced in live code. The headline recommendation: do NOT make any public "before it arrives" lead claim until (a) a non-circular Wikipedia+GDELT no-gap anchor is wired as a held-out validator and the mainstream-collapse + gap-incoherence bugs are fixed, and (b) the accuracy ledger clears its product-success bar (the moat charter's own bar is currently CONTRADICTED at 0.0% honest hit rate).

The instrument is promising and structurally honest in its design, but it is not yet a defensible, ledger-backed "human attention" instrument across the full diffusion curve. Caveat on scope: this audit rests on a 100-topic no-gap sample within a 2,103-topic snapshot and an 88-instrument market pull, all single-day (2026-06-22/23) - directionally decisive but not a multi-week out-of-sample validation.

1. What the Gradient Score Is

The Gradient Score is an "attention intelligence" instrument that emits TWO headline numbers from ONE measurement set, built on an explicit (not metaphorical) Heisenberg trade-off: a trend's POSITION on the diffusion curve (how early/niche it is) and its MOMENTUM (whether it will sustain) cannot both be measured at maximum precision at once, because waiting to confirm momentum is the act of measurement that destroys earliness. DETECTION is the earliness-optimized score (accepts ~22% false positives, audience: creators/marketers);

CONFIDENCE is the precision-optimized score (target <9% FP, audience: institutional analysts/investors). The GAP = Detection - Confidence is treated as the single most precise read of where a topic sits on the curve: a large positive gap = detected early, not yet broadly confirmed; the gap closes UPWARD as a genuine trend matures (both converge HIGH = "confirmed, act now") and closes DOWNWARD as a false positive collapses (both fall LOW).

Displayed STAGES are Detection-derived with exact cutoffs: BREAKOUT ≥ 85 , STRONG 70-84, EMERGING 55-69 (shown to users as "INDICATING" - display-only rename, internal key stays EMERGING), MARGINAL/WATCHING 35-54, MONITORING < 35 ; ANOMALY is an orthogonal flag, not a strength band, firing when Detection runs 16+ points ahead of Confidence.

A THIRD score, N (nowtrendin_score), measures internal in-app query demand and is the ranking key for the "Now TrendIn" view (while "All Signals" ranks by Detection); critically, N is EXCLUDED from the Gradient Score itself to prevent a reflexive search->score->search feedback loop. DOC-vs-CODE reconciliation, with drift flagged: (1) the docs/CLAUDE.md and even in-code comments still cite headline weights of G 0.40 / D 0.25 / I 0.20 / M 0.10 / C 0.05 for Detection and an N weight of 10-12% - these are STALE; live code runs a six-component (G/I/M/D/C/P, plus Persistence) N-excluded set renormalized to 1.0, with N at 0.0 everywhere.

(2) METHODOLOGY.md names the displayed bands "BREAKOUT/STRONG/EMERGING/WATCHING/MONITORING" while signals.ts uses MARGINAL as the primary 35-54 label and renames EMERGING->INDICATING - a minor doc lag. (3) "Overall" is defined two different ways: weighted-six at scoring time vs (Detection+Confidence)/2 at serve/calibration time, so /scores and /detail can diverge.

(4) The legacy five-component gradient_engine_backend.py (no P, no N) is NOT the serve path and must not be cited as authoritative; the canonical live scorer is gravitational_anomaly_detector.py.

Displayed stage cutoffs (Detection-derived)

Stage (internal key)	User label	Detection cutoff	Meaning
BREAKOUT	≥ 85	Breakout	Arrived / confirmed, act now
STRONG	70 - 84	Strong	Strong, broadly corroborated
EMERGING	55 - 69	INDICATING	Detected, not yet broadly confirmed
MARGINAL / WATCHING	35 - 54	Marginal	Weak / watch
MONITORING	< 35	Monitoring	Below signal floor
ANOMALY (flag)	Det 16+ over Conf	Anomaly	Orthogonal flag, not a strength band

2. Data Factors and How They Are Weighed

Live code weights (six-component, N excluded, renormalized to 1.0)

Component	Detection	Confidence	Overall	What it measures
G - Gradient Strength	0.375	0.122	0.244	Niche-to-mainstream engagement density (fires earliest)
D - Dark Matter	0.216	0.044	0.133	Inferred hidden/private-conversation signal (first-timer ratio, asymmetry)
I - Inertia	0.182	0.278	0.222	Consecutive accelerating 6h windows (best sustain predictor)
M - Medium/Platform	0.102	0.222	0.167	Cross-platform diffusion breadth
C - Confidence Decay	0.057	0.067	0.078	Freshness / age (Heisenberg-exempt)
P - Persistence	0.068	0.267	0.156	Cycles held above EMERGING (self-referential - see note)

Detection leads on the earliest-firing components (G + D); Confidence leads on the slowest-to-confirm (I + P + M). N (in-app demand) is verifiably 0.0 in all three composites.

The score is built from six external measurement components, each scored 0-100. INTENT (May-2026 Heisenberg PDF, faithfully built in the legacy 5-component module): G - Gradient Strength (niche-to-mainstream engagement density ratio; fires instantly from ~8 signals);

I - Inertia (consecutive 6h windows of accelerating signal count; highest-latency, best sustain predictor); M - Medium/Platform diversity (cross-platform diffusion pattern match);

D - Dark Matter (probabilistic hidden-private-conversation inference from first-timer ratio, engagement asymmetry, vocabulary clustering); C - Confidence Decay (freshness/age, Heisenberg-exempt).

The live canonical engine adds P - Persistence (cycles held above EMERGING). LIVE CODE WEIGHTS (renormalized six-component, N=0): Detection = G 0.375 + D 0.216 + I 0.182 + M 0.102 + C 0.057 + P 0.068;

Confidence = I 0.278 + P 0.267 + M 0.222 + G 0.122 + C 0.067 + D 0.044; Overall = G 0.244 + I 0.222 + M 0.167 + D 0.133 + C 0.078 + P 0.156.

The structure is faithful to intent (Detection leads on the earliest-firing G+D; Confidence leads on the slowest-to-confirm I+P+M), but WEIGHT DRIFT is real and material: published/comment weights (G 0.40, N 10-12%) do not match live constants, and the in-code comment block at apply_calibration lines 1100-1101 lists weights that match NEITHER the docs NOR the code below it.

DUAL-PATHWAY: every topic is scored on the expert G / I / M / D / C / P composite by default, then a dual-pathway blend runs last with weight $w = \text{mainstream_ratio}$ (baseline-relative deviation from the topic's OWN 12-cycle footprint). Mainstream-tier evidence lands in G's DENOMINATOR (suppresses the early read) and raises w ; expert/niche tier keeps $w \sim 0$ and preserves the expert gradient (the moat).

This is the correct design - fame != movement - and it visibly protects the top tier (FIFA would otherwise collapse to 27.6). WHERE WEIGHTING IS SUSPECT: (1) the same-class ESTABLISHED discount has THREE different multipliers depending on maturity source (0.25 / 0.40 / 0.45) - a calibration inconsistency.

(2) Overall and Stage are computed on different inputs at scoring vs serve time, so a topic can carry two different stages. (3) The mainstream-collapse evidence (see error analysis) shows the dual-pathway/calibration preservation is NOT firing for mid-mainstream rows.

CIRCULARITY ASSESSMENT: the headline integrity rule is upheld - N (in-app demand) is verifiably excluded from all three composites, so there is NO forbidden internal-feedback leak; ranking-by-N is a display choice that does not contaminate the score. Dark Matter uses external-platform (Reddit) first-timer/asymmetry signals, NOT NowTrendIn's own users - acceptable.

The ONE real concern: PERSISTENCE is self-referential - it is derived from topic_lifecycle counters built from the engine's own prior OVERALL scores crossing the EMERGING threshold, and P then feeds Overall (0.156) and Confidence (0.267). Inputs G/I/M/D/C are independent, so it is feedback rather than pure circularity, but P is NOT an input independent of what it validates - the classic violation of the project's own "validator must use independent inputs" standard, and a standing item to watch.

3. Distribution, Saturation, and Separation

Per-stage distribution across the 2,103-topic live snapshot

Stage	Count	Det median	Conf median	Gap median	Det range	Conf range
BREAKOUT	10	90.5	91	-0.8	84.5 to 94.7	81.8 to 94.3
STRONG	41	74.1	80.3	-8.2	40.0 to 91.8	48.0 to 88.1
EMERGING	106	46	66.8	-21.6	40.0 to 74.5	48.0 to 76.9
WATCHING	762	32.8	51.7	-22.4	19.3 to 100.0	34.6 to 100.0
MONITORING	1184	17.3	25.3	-7.9	3.8 to 100.0	7.0 to 97.5
ANOMALY (flag)	291	35.3	47.7	-16.5	14.0 to 100.0	15.5 to 100.0

Reads as: the score is NOT collapsed - stage medians ladder monotonically (Det 90.5 / 74.1 / 46.0 / 32.8 / 17.3) and 520 of 595 Detection/Confidence values are distinct. But note the inversion in the middle of the curve: at EMERGING and WATCHING the **gap median is ~-22** (Confidence well above Detection) - the opposite of the intended "early = Detection leads." The gap converges to ~0 only for genuinely-arrived BREAKOUT topics (median -0.8).

4. Critical Error Analysis

4a. The no-gap falsification test

This is where the instrument is weakest. THE DECISIVE FAILURE (basis: WebSearch-confirmed against the Federal Reserve official statement + CBS/NBC/ABC): Alan Greenspan died 2026-06-22 at 100 - the day's #1 confirmed mainstream story - and scored Detection 91.8 / Confidence 75.6 / gap +16.2, the sole large positive gap in the sample and squarely in the Anomalies band.

The no-gap principle explicitly defines this exact case (an obviously-mainstream, fully-confirmed topic read as early/uncertain) as the broken state that invalidates the gap metric. The likely mechanism: Detection is correctly high, but Persistence has not accrued for a one-day-old event, depressing Confidence - the documented "gap closes as P rises" lifecycle - but for a wall-to-wall confirmed obituary the instrument must not read it as a speculative lead.

This needs a same-surge/breaking-news check. THE SYSTEMATIC FAILURE (basis: internal distribution + footprint counts): seven mainstream nations/mega-caps collapse to a near-identical signature - china 45.6/67.5 (219 mentions/11 platforms), india 45.6/67.5, russia 45.5/67.3, france 45.3/67.5, apple 45.2/67.2, openai 45.4/67.1, korea 45.4/67.1 - all badged EMERGING with a uniform ~-22 LAGGING gap; obama is worse (24.3/56.7, WATCHING, gap -32.4).

Confidence systematically exceeds Detection on already-arrived topics, the inverse of the intended design, indicating Detection is being clamped below what the footprint justifies and that the dual-pathway preservation gate is not firing for mid-mainstream rows. CONFIRMED ERROR (adversarially verified, severity medium): "strait of hormuz" reads Detection 23.9 / WATCHING while the shorter "hormuz" token reads STRONG 72.3/83.2 - the SAME story (the dominant 2026 oil/geopolitics flashpoint, IEA-described as the largest supply disruption in oil-market history).

Root cause verified in code: `normalize_topic()` only lowercases/strips punctuation and there is no alias-consolidation layer for non-AI topics, so surface-form variants are scored as independent topics; the canonical full phrase accumulates fewer hits and collapses. CONFIRMED ERROR (verified, severity low): "belgium vs iran" (a real 6/21 World Cup fixture) reads 10.2/17.7 MONITORING - under-weight but explainable by a known, DOCUMENTED, fix-pending mechanism (CLAUDE.md sec 15 M/D reweighting names this exact case as IN DESIGN).

Note: the auditor's proposed "wide positive gap" expectation here is wrong - a fully-arrived mainstream event should read HIGH on both axes, not with a wide gap; the auditor conflated mainstream with early-emerging. DATA-INTEGRITY BUG (basis: arithmetic): the stored `heisenberg_gap` contradicts Detection-Confidence on the entire mcp family (mcp 74.5-73.0 = +1.5 but stored gap -23; same on claude code, claude-code, model-context-protocol, mcp server) - a ~24pt internal contradiction breaking the "both scores explain each other" test, independent of the mainstream question.

ACCURATE EXAMPLES (the engine genuinely works here): trump 94.6/94.2/+0.3, FIFA 93.7/93.7/0.0, iran 86.1/90.1/-4, election 94.7/94.3/+0.4, canada 90/91.7/-1.7, claude 77.8/86.2 STRONG - all high+high+aligned, the correct already-arrived read. NEGATIVE CONTROLS also pass: SpaceX IPO 2026 (24.9/25.4, n=0) and obscure dev-tool slugs (fastapi 46.2/48.6, langchain 45.5/47.9) sit LOW on BOTH axes - the engine does NOT fabricate confidence for niche topics, so the failure is ONE-DIRECTIONAL (it strips score from mid-mainstream topics; it does not invent it for niche ones).

SATURATION/SEPARATION (basis: 2,103-topic distribution): the instrument is NOT collapsed - stage medians ladder monotonically (Detection 90.5/74.1/46.0/32.8/17.3), 520/595 distinct Detection/Confidence values exist, and the gap converges to ~0 only for genuinely-arrived BREAKOUT topics (median -0.8). Real caveats: heavy near-duplicate string variants score identically (inflating apparent compression and the catch-all), some single-platform GitHub slugs carry mid-range Detection on one source, and a few thin BREAKOUT/anomaly rows ("bonkers", 5 mentions/4 platforms, 86-91 Detection) warrant a corroboration check.

HONEST SAMPLE/AGE CAVEAT: all of the above is a single-day snapshot (2026-06-22) over a 100-topic hand-checked sample within 2,103 topics; it is directionally decisive (it fails on the day's #1 event plus a reproducible 7-topic pattern) but is not a multi-week out-of-sample false-positive measurement, and the stated 22%/<9% FP rates remain design targets, not ledger-confirmed.

4b. The mainstream-collapse signature (live sample)

Already-arrived nations and mega-caps collapse to a near-identical ~45 Det / ~67 Conf EMERGING signature with a uniform ~-22 gap - Confidence exceeding Detection on topics that have unambiguously arrived:

Topic	Stage	Detection	Confidence	Gap	Mentions	Platforms
china	EMERGING	45.6	67.5	-21.8	219	11
india	EMERGING	45.6	67.5	-21.8	156	11
russia	EMERGING	45.5	67.3	-21.8	84	9
france	EMERGING	45.3	67.5	-22.1	94	11
apple	EMERGING	45.2	67.2	-22	72	8
openai	EMERGING	45.4	67.1	-21.7	77	7
korea	EMERGING	45.4	67.1	-21.8	79	7
obama	WATCHING	24.3	56.7	-32.4	51	9

4c. Two adversarially-confirmed errors + the decisive event

Case	Det	Conf	Gap / Stage	Finding
Alan Greenspan (died 6/22, the day's #1 story)	91.8	75.6	+16.2 / STRONG	Largest positive gap in the sample; a confirmed obituary read as a speculative early lead - the exact broken state the no-gap principle names.
'hormuz' vs 'strait of hormuz' (same story)	72.3 / 23.9	83.2 / 54.3	STRONG vs WATCHING	CONFIRMED ERROR (medium): surface-form fragmentation - no alias-consolidation layer, so the canonical phrase collapses to WATCHING while the short token reads STRONG.
'belgium vs iran' (real 6/21 WC fixture)	10.2	17.7	MONITORING	CONFIRMED ERROR (low): under-weight, but explainable by the documented, fix-pending M/D reweighting (CLAUDE.md sec 15).
mcp family (stored heisenberg_gap)	74.5	73.0	stored gap -23	DATA-INTEGRITY BUG: stored gap contradicts Det-Conf (=+1.5) by ~24 pts - breaks the 'both scores explain each other' test.

4d. Where the engine is genuinely correct (accurate examples)

Topic	Det	Conf	Gap	Why it is correct
trump	94.6	94.2	0.3	The textbook PASS of the no-gap test: the single most-covered person on earth (609 mentions / 13 platforms) reads det 94.6 / conf 94.2 / gap +0.3 - bo
iran	86.1	90.1	-4	Central to the active war story; 542 mentions / 12 platforms. det 86.1 / conf 90.1 / gap -4 - both high, tightly aligned, BREAKOUT. Correctly pegs an
FIFA	93.7	93.7	0	World Cup is live and globally dominant; det 93.7 / conf 93.7 / gap 0.0 - exact convergence. The dual-pathway anti-collapse gate is visibly working he
election	94.7	94.3	0.4	det 94.7 / conf 94.3 / gap +0.4, BREAKOUT - high+high+aligned for a mainstream civic topic.
Keir Starmer	77.6	84.3	-6.7	Sitting UK PM; 326 mentions / 12 platforms. det 77.6 / conf 84.3 / gap -6.7 - both high, modest lag, STRONG. Reasonable already-arrived read.
canada	90	91.7	-1.7	det 90 / conf 91.7 / gap -1.7, BREAKOUT - mainstream nation read high+high+aligned. Correct.

Topic	Det	Conf	Gap	Why it is correct
claude	77.8	86.2	-8.5	Major AI brand; det 77.8 / conf 86.2 / gap -8.5, STRONG - both high, coherent. Correct already-arrived read.
SpaceX IPO 2026	24.9	25.4	-10.5	A genuinely speculative/forward-looking topic (n=0, no baseline). Low det + low conf + MONITORING is the CORRECT 'false-positive collapse' behavior -

The top tier passes cleanly (trump 94.6/94.2, FIFA 93.7/93.7, election 94.7/94.3, canada 90/91.7) and niche topics correctly sit LOW on BOTH axes (SpaceX-IPO 24.9/25.4, fastapi 46.2/48.6) - the engine does not fabricate confidence. The failure is one-directional: it strips score from mid-mainstream topics; it does not invent it for niche ones.

5. Market Signal Categories

Tier distribution across the live 88-instrument universe

Tier	Count	% of universe
ROUTINE	87	98.9%
BUILDING	1	1.1%
ELEVATED	0	0.0%
ACTIVE	0	0.0%
DORMANT	0	0.0%
WATCH	0	0.0%
calibrating=true (overlay)	25	28.4%

87 of 88 instruments (98.9%) are ROUTINE - the user-facing tier carries essentially zero information. The baseline-relative z-score design is genuinely good (it fixes the mega-cap size-bias) and cold-start honesty is real, but as surfaced the tiers do not separate instruments, and three tier fields (tier / risk_stage / classification) disagree on the same row.

The Market Signal tier ladder (ELEVATED ≥ 80 / ACTIVE 60-80 / BUILDING 40-60 / ROUTINE 25-40 / DORMANT < 25 , on the Detection+Confidence average, with CALIBRATING overriding) is conceptually sound and the engine's baseline-relative design is genuinely strong: every component is scored as a z-score against the item's OWN history ($z=0 \rightarrow 0.30$, $z=2 \rightarrow -0.74$), which correctly fixes the size-bias bug where mega-caps auto-scored ELEVATED on coverage volume alone. The CALIBRATING short-circuit is well-implemented and hard to bypass (a fresh IPO like SPCX trips all components and is honestly flagged "snapshot, not a confirmed signal"), which is exactly the cold-start honesty an institutional buyer should want.

BUT as surfaced today the tier system does NOT track or separate instruments coherently. From the live 88-instrument pull (all fresh, scored within 0.7h - staleness is NOT the problem): the headline tier (market_gradient.tier / risk_stage) is ROUTINE for 87/88 (98.9%) - only alphabet is BUILDING - so the user-facing tier carries essentially ZERO information.

THREE tier fields disagree on the same row: tier vs risk_stage vs classification (e.g. cre = ROUTINE tier but ELEVATED classification; recession, spacex, jpmorgan = ROUTINE tier but WATCH classification). There is no single source of truth.

COLD-START CONTRADICTION: 28.4% of instruments are flagged calibrating=true, yet 0% have sufficient_baseline=false - every calibrating row simultaneously reports "enough baseline." The two gates are computed on different inputs (calibrating ~ thin total_signals 1-6; sufficient_baseline ~ cycle count) and are never reconciled. VALUE QUANTIZATION: Detection snaps to a few discrete buckets (20 instruments at 34.6, 16 at 32.2, 9 at 33.9), and calibrating rows collapse to floor pairs (29.9/29.9, sig=1) presented in the same shape as measured reads.

ABNORMALITY IS NOISE-DOMINATED on thin coverage: the nine "highest abnormality" instruments (abn 4.8, delta 5.3%) are all 2-signal micro-cap floats (Lakewood Amedex, Boxlight, Utime) classed ROUTINE, while jpmorgan (abn 2.4, sig 125) is WATCH - so the "biggest movers" sort surfaces illiquid noise above real macro movers (recession, spacex). STRUCTURAL COVERAGE GAP: the tradable universe is a hardcoded ~16-name watchlist (15 mega-caps + SpaceX); anything off it silently zeroes positioning/fundamentals/momentum - the "SpaceX problem." PARITY RISK: two divergent market formulas coexist (the z-score engine that is actually persisted vs an absolute-magnitude inline variant tied to the Grade tool), so the Market tab and Grade tool can disagree for the same ticker - a CLAUDE.md 3-platform-parity violation.

Also flagged: a hardcoded identical interpretation string across all instruments and double-encoded UTF-8 mojibake in the payload (will render as garbage on the terminal). NET: the engine is directionally honest about a quiet tape and its baseline-relativity is real, but the tiers as surfaced are uninformative and internally contradictory; "mostly ROUTINE" is defensible for a quiet market but is not the same as a tier system that is tracked accurately and surfaced coherently.

6. External Benchmarks and a Non-Circular Calibration Loop

Candidate external attention anchors

Benchmark	Measures	Cost / licensing	Non-circular?	Use
Wikipedia / Wikimedia Pageviews API (AQS REST	Daily (and hourly/monthly) human pageviews to a specific Wikipedia article - i.e. how many real people actively looked u	FREE. Data is licensed CC0 1.0 (public domain) - the stronge	FULLY non-circular. Pageviews come from Wikimedia's own serv	WIRE AS THE NO-GAP MAINSTREAM ANCHOR. (1) For each scored topic, resolve it to a canonical
GDELT DOC 2.0 API (TimelineVol mode)	Share of WORLDWIDE news coverage (broadcast + print + web, 100+ languages) matching a query, by day/hour/15-min - expres	FREE. DOC 2.0 API is open, no key, JSON/CSV/JSONP output, ro	Largely non-circular but needs care. GDELT monitors a far br	SECONDARY MAINSTREAM CONFIRMATION. Use TimelineVol percent-coverage as a second independen
Google Trends API (alpha, July 2025) / relativ	Relative search interest for a term on a consistent 0-scaled index across requests (5 years of history in the alpha). Se	Official API is FREE but ALPHA + gated - access opened on a	PARTIALLY circular for NowTrendIn - IMPORTANT CAVEAT. The ac	KEEP IN ITS CURRENT LANE (breakout timing), do NOT reuse for no-gap. If you want a search-
GoogleBooks / Ngram Viewer (arrival-lag anchor	Frequency of a term across Google's scanned-books corpus by year - culturomics. Captures when a term entered the durable	FREE. Dataset is CC BY 3.0 (attribution required, commercial	Non-circular (books corpus shares nothing with our feeds) BU	RETROSPECTIVE THESIS VALIDATION ONLY. For a basket of historical topics, measure detection
Reddit API (community attention)	Post/comment volume and subreddit activity for a topic - grassroots/community attention, often an early-signal layer tha	PAID + opaque. Commercial access ~ \$12,000/month for up to 5	Non-circular IF NowTrendIn does not already ingest Reddit (i	OPTIONAL early-signal cross-check, not a mainstream anchor. Given cost + opacity and that
X / Twitter API (real-time social attention)	Real-time post volume/engagement for a topic - social-media attention velocity.	PAID + expensive. Pay-per-use default (\$0.005/read, capped 2	CIRCULAR for NowTrendIn - DISQUALIFIED as an anchor. The eng	DO NOT USE as a calibration anchor - it would validate the score with the score's own input
Media-monitoring indices (MIT/Harvard Media CI	Media Cloud: open-source index of millions of news stories, story-count attention over time per topic - similar construc	Media Cloud = FREE / open (academic, API access). Commercial	Media Cloud is non-circular and independent; its weakness is	Use Media Cloud as a FREE swap-in/redundancy check for GDELT on the media-supply axis. Avo

The no-gap test currently has no objective ground truth - it was scored by hand against WebSearch. The fix is a free, non-circular external anchor, and the clear winner is the WIKIPEDIA / WIKIMEDIA PAGEVIEWS API (AQS REST v1).

It is the textbook attention ground-truth in published research, it is genuinely FREE, it is licensed CC0 1.0 (public domain - the cleanest possible position for a commercial product to ingest and even resurface), and it is FULLY non-circular: pageviews come from Wikimedia's own server logs, an independent population on a platform NowTrendIn does not collect from, sharing ZERO inputs with the Gradient Score (no RSS, no X scan, no blog feeds, no Detection/Confidence components). For the no-gap test it maps almost directly: an already-mainstream topic reliably has high, STABLE article traffic; a niche/pre-arrival topic has thin or no traffic.

Pair it with GDELT DOC 2.0 (TimelineVol, free, percent-of-global-coverage) as an independent MEDIA-SUPPLY second axis - true mainstream should be HIGH on BOTH the demand axis (Wikipedia) and the supply axis (GDELT); disagreement is itself diagnostic (high GDELT + low Wikipedia = media-pushed, no public pull). Two critical exclusions on integrity grounds: X/Twitter is DISQUALIFIED as an anchor because it is already a scoring INPUT (the 6h scan) - using it as ground truth is the exact circular-metric failure the integrity standard forbids; and Google Trends must stay in its EXISTING lane as the breakout/arrival-timing anchor in the accuracy ledger - reusing it for no-gap would double-anchor to the same provider's attention definition and over-fit.

Reddit (paid ~\$12k/mo, opaque, and the wrong/early construct) and Google Ngram (annual, years-lagged, retrospective-only) are not live anchors. CONCRETE NON-CIRCULAR CALIBRATION LOOP (weekly, mirroring the accuracy-ledger cadence): STEP 0 - hard firewall: Wikipedia/GDELT data is written ONLY to a dedicated held-out table (attention_anchor) that NO Gradient-Score component, calibration weight, dual-pathway gate, or serve_payload ever reads.

STEP 1 - resolve each scored topic to a canonical Wikipedia article (title-normalize + search; unresolved topics QUARANTINE to a review queue, never guessed - same condition-precedent discipline as the ingestion gate; this also directly attacks the surface-form fragmentation bug). STEP 2 - pull trailing ~90d daily pageviews + GDELT coverage-share; compute mainstream-LEVEL = $\log_{10}(\text{median daily views})$, STABILITY = low coefficient of variation, and coverage-share level.

STEP 3 - label ALREADY-MAINSTREAM = pageviews above a high corpus percentile (~p80) AND stable AND high GDELT, EARLY = low+rising. STEP 4 - score the no-gap test as a confusion matrix; the key failure metric is FALSE-EARLY (we called something emerging that the anchors say is already mainstream - precisely the Greenspan/country-collapse failure).

STEP 5 - calibrate DIRECTION only: a rising false-early rate prompts a HUMAN-reviewed adjustment of the SATURATION/ESTABLISHED-discount thresholds; the external number is never regressed into a weight. STEP 6 - a WORSENING drift alarm (like catchall_floor_log) trips when false-early climbs past a set delta.

This stays permanently non-circular because the anchors live in a held-out table the score cannot read, share no inputs with the score, are never auto-fit, and are kept on a different external series from the breakout (Trends) job.

7. Recommendations

Quick wins - shippable now, no research gate, each integrity-preserving

- FIX the heisenberg_gap data-integrity bug - the stored gap must equal Detection-Confidence on every row; the mcp-family ~24pt contradiction is a flat arithmetic/storage bug and the easiest credibility win. Pin down whether the signed-vs-abs() divergence between scoring-time (signed) and apply_calibration (abs) is the culprit and standardize on signed.
- ADD a surface-form alias/consolidation layer to normalize_topic()/extract_topics() so "hormuz / strait of hormuz / strait hormuz" and "mcp / model-context-protocol / mcp server" collapse to one canonical topic before scoring - fixes both the strait-of-Hormuz error and the duplicate-driven apparent saturation.
- RECONCILE the three market tier fields to ONE surfaced source of truth, and reconcile the calibrating vs sufficient_baseline contradiction (gate both on the same definition).
- GATE market abnormality on a minimum signal count so 2-signal micro-floats stop topping the mover ranking.
- FIX the UTF-8 double-encoding mojibake in /risk/scores and replace the hardcoded identical interpretation string with a per-instrument read or remove it.
- RECONCILE the stale weight comments (apply_calibration lines 1100-1101) and CLAUDE.md/docstrings to the actual live six-component weights, and align METHODOLOGY.md band names (MARGINAL/INDICATING) - documentation honesty.
- RESOLVE the two divergent market formulas (z-engine vs absolute-magnitude inline) to a single persisted engine so the Market tab and Grade tool agree (parity rule).

Research-gated - require backtest-before-ship, must not bypass integrity

- BUILD the breaking-news / same-surge override for the no-gap top end so a high-Detection, confirmed, wall-to-wall event (Greenspan) is not read as a speculative lead - this likely means letting Detection-and-corroboration, not slow-accruing Persistence, drive Confidence for confirmed mega-events; validate it does not re-open false positives.
- EXTEND the dual-pathway anti-collapse preservation across the full mainstream band so china/india/apple/openai stop collapsing to the ~45/67 EMERGING signature - diagnose whether their baseline-relative w is falling too low or the serve-time preservation gate is mis-firing; gate behind the new Wikipedia/GDELT no-gap precision report.
- WIRE the Wikipedia+GDELT held-out no-gap calibration loop (Section above) BEFORE any public no-gap claim.
- ADDRESS the Persistence self-reference: either down-weight P or add an independent persistence proxy, since it currently feeds on the engine's own past output (integrity-standard concern).
- EXPAND the market universe beyond the 16-name watchlist via the market-coverage skill so trending real tickers aren't silently zeroed.
- DO NOT publish any "before it arrives" lead figure until the accuracy ledger clears its own product-success bar (median lead $\geq +3d$, LED $\geq 35\%$, FP $\leq 30\%$, ≥ 100 matured events) - it is currently CONTRADICTED at 0.0% honest hit rate, and the moat charter itself requires printing FAIL until the denominator supports the claim.

8. Verdict

Not yet - NowTrendIn cannot today make an unqualified "measures where human attention is moving before it arrives" claim to an institutional buyer, because the instrument fails its own no-gap credibility floor on the day's single most important confirmed event (Greenspan, +16.2 anomaly gap) and on a systematic class of already-arrived mainstream topics that collapse to a ~45/67 EMERGING signature, while the moat's lead claim is currently UNPROVEN and at last ledger CONTRADICTED (0.0% honest hit rate). What the product CAN defend right now is real and not trivial: a structurally honest dual-score design faithfully implemented in code, a verified no-N-into-Gradient firewall (the core integrity rule), genuinely baseline-relative market scoring with disciplined cold-start honesty, clean monotonic separation across 2,103 topics, demonstrably correct already-arrived reads for the famous top tier (trump/FIFA/iran/election), and a one-directional failure mode that never fabricates confidence for niche topics.

For this to become a defensible "human attention" instrument, the following must be true: the heisenberg_gap and surface-form-fragmentation bugs are fixed; the dual-pathway anti-collapse protection extends across the full mainstream band; a held-out, non-circular Wikipedia+GDELT no-gap anchor is wired and shows acceptable false-early precision; the market tier fields and cold-start flags are reconciled into a single coherent surface; and the accuracy ledger clears its own published success bar so the "before it arrives" lead is denominator-backed rather than a claim. Until then it is a promising, honestly-architected pre-validation instrument - sellable as a measurement tool with explicit caveats, but not yet as a proven attention-prediction moat.

What must be true before the "before it arrives" claim is defensible to an institutional buyer: (1) the heisenberg_gap and surface-form fragmentation bugs are fixed; (2) the dual-pathway anti-collapse protection extends across the full mainstream band; (3) a held-out, non-circular Wikipedia + GDELT no-gap anchor is wired and shows acceptable false-early precision; (4) the market tier fields and cold-start flags are reconciled into one coherent surface; (5) the accuracy ledger clears its own published success bar so the lead is denominator-backed.

Appendix A - 100-topic no-gap sample (live engine, 2026-06-22/23)

Stratified across stages. Gap = Detection - Confidence. Sorted by Detection desc.

Topic	Stage	Det	Conf	Gap	Ment.	Plat.	N
asi	BREAKOUT	94.7	94.3	0.4	143	12	76.6
election	BREAKOUT	94.7	94.3	0.4	85	11	76.8
trump	BREAKOUT	94.6	94.2	0.3	609	13	78.8
FIFA	BREAKOUT	93.7	93.7	0	62	11	72.4
Alan Greenspan	STRONG	91.8	75.6	16.2	59	7	75.4
bonkers	BREAKOUT	91	81.8	9.2	5	4	54.4
canada	BREAKOUT	90	91.7	-1.7	62	10	78.3
africa	BREAKOUT	86.5	90.2	-3.8	44	9	89.1
iran	BREAKOUT	86.1	90.1	-4	542	12	85.4
qatar	BREAKOUT	85.6	89.5	-3.9	60	11	83.1
fifa world	BREAKOUT	84.5	89.3	-4.8	39	8	100
japan	STRONG	81.8	88.1	-6.3	111	9	78.7
america	STRONG	80	87.3	-7.2	98	10	86.2
europa	STRONG	80	87.2	-7.2	96	9	78.2
ukraine	STRONG	79.1	86.8	-7.8	110	11	83.1
anthropic	STRONG	77.9	86.1	-8.2	98	9	73.6
claude	STRONG	77.8	86.2	-8.5	179	11	73.5
Keir Starmer	STRONG	77.6	84.3	-6.7	326	12	78.4
mcp	EMERGING	74.5	73	-23	70	3	81
claude code	WATCHING	74.5	67.8	-17.1	86	2	78
claude-code	WATCHING	74.5	67.8	-22.4	68	1	51
model-context-protocol	MONITORING	74.5	67.8	-2.8	33	1	51
mcp server	WATCHING	74.5	67.8	-23.5	30	2	83.5
agi	STRONG	72.6	83.3	-10.8	75	11	78.3
hormuz	STRONG	72.3	83.2	-10.9	128	11	77.7
israel	STRONG	72	82.7	-10.8	169	10	81.7
australia	STRONG	71.6	80.3	-8.7	110	11	83.8
britain	STRONG	70.7	81.9	-11.3	103	9	99.4
california	STRONG	67.4	79.9	-12.5	74	7	76.5
rag	STRONG	65	60	-7.8	316	12	72.8
germany	EMERGING	62.4	76.9	-14.4	91	10	79.7
spacex	EMERGING	60.2	76.1	-15.8	132	10	84.5
iran war	EMERGING	60.2	75.8	-15.6	73	8	84.9
ipo	EMERGING	58.6	60.2	-1.5	69	7	75.6
vance	EMERGING	57.4	53.3	4.1	105	9	73.7
strait	EMERGING	57	74.8	-17.8	93	11	80.3
ai agents	EMERGING	50.8	60	-24.3	113	7	85
llm	EMERGING	50.7	60	-24.4	342	7	83.2

Topic	Stage	Det	Conf	Gap	Ment.	Plat.	N
ai agent	EMERGING	48.8	60	-25.6	146	6	85
burnham	WATCHING	48.6	48.4	0.3	142	9	78
keir	WATCHING	47.9	47.8	0.1	140	9	58.1
andy burnham	WATCHING	47.7	47.7	0	91	8	97.8
typescript	WATCHING	47.1	54.5	-17.5	43	2	76.9
cursor	WATCHING	46.6	54.6	-18.1	34	2	78
fastapi	WATCHING	46.2	48.6	-13.8	52	1	54.4
us-iran	EMERGING	46	68.4	-22.4	116	8	88.9
world cup 2026	EMERGING	46	68	-22	68	8	89.8
england	EMERGING	45.9	68	-22.1	82	9	74.3
prompt-engineering	WATCHING	45.7	56.5	-22	89	2	66.4
ai tools	WATCHING	45.7	48.5	-14.2	62	1	97
china	EMERGING	45.6	67.5	-21.8	219	11	72.4
india	EMERGING	45.6	67.5	-21.8	156	11	84.8
codex	WATCHING	45.6	48.3	-14	38	1	92.3
russia	EMERGING	45.5	67.3	-21.8	84	9	79
langchain	WATCHING	45.5	47.9	-13.8	40	1	57.2
phi	EMERGING	45.4	67.4	-22	103	10	80.3
korea	EMERGING	45.4	67.1	-21.8	79	7	80.8
openai	EMERGING	45.4	67.1	-21.7	77	7	72.8
france	EMERGING	45.3	67.5	-22.1	94	11	88.9
apple	EMERGING	45.2	67.2	-22	72	8	76.5
agentic ai	WATCHING	44.2	60	-23.5	39	4	80.4
ollama	WATCHING	43.7	47.8	-14.2	34	1	52.9
agentic	WATCHING	43.5	60	-29.1	54	4	87.2
embeddings	WATCHING	43.2	47.4	-14.3	63	1	57.2
ai safety	WATCHING	41.6	58.2	-21.8	59	3	78
gemini	WATCHING	41.5	60.9	-25.9	43	5	85
generative ai	EMERGING	40	48	-22.8	78	4	79.7
colombia	WATCHING	38.5	42.2	-3.8	69	8	72.8
agent	WATCHING	38.3	63.4	-25.1	302	6	75.4
ai-safety	WATCHING	37.5	46.7	-20.6	56	1	76.2
vector-database	WATCHING	36.8	46.6	-21	54	1	46.7
machine-learning	WATCHING	36.5	48	-27.6	49	2	89.3
ai-agents	WATCHING	35.8	47.3	-22.7	87	1	51
developer-tools	WATCHING	35.7	45.7	-21.4	67	1	73
ai-tools	WATCHING	35.5	45.8	-21.7	67	1	51
llama	WATCHING	32.6	57.1	-24.5	57	4	88.1
embedding	WATCHING	32.2	44.7	-22.6	67	1	80
generative-ai	MONITORING	30.9	21.1	-1.4	67	1	46.7
fine tuning	WATCHING	30.5	49	-18.4	27	2	75.4

Topic	Stage	Det	Conf	Gap	Ment.	Plat.	N
agent ai	MONITORING	29.8	19.6	-1.2	33	1	46.7
ai-agent	MONITORING	29	20.7	-2.9	92	1	66.4
agents ai	MONITORING	28.6	19.7	-2.4	51	1	76.9
us open	WATCHING	25.8	52.1	-26.3	56	5	78.5
python rag	MONITORING	25.6	43.6	-24.2	51	1	82.1
SpaceX IPO 2026	MONITORING	24.9	25.4	-10.5	68	3	0
switzerland	WATCHING	24.8	57.8	-33	104	9	82.8
strait hormuz	WATCHING	24.7	56.7	-32	92	11	76.2
lebanon	WATCHING	24.4	57.1	-32.7	88	10	79.2
obama	WATCHING	24.3	56.7	-32.4	51	9	85
strait of hormuz	WATCHING	23.9	54.3	-30.4	75	11	70.4
zelensky	MONITORING	23.6	33.8	-10.1	30	9	62.5
agentic-ai	MONITORING	20.9	18	-3.3	30	1	46.7
fine-tuning	MONITORING	19.2	40.8	-21.4	34	1	99.1
meloni	MONITORING	18.4	30.8	-12.4	29	8	84.5
warsh	MONITORING	15.6	27.1	-11.5	38	3	57.1
wyndham	MONITORING	14.7	26.5	-11.8	31	4	91.3
Deniz Undav	MONITORING	13.5	25.5	-12	63	7	68.5
the korea times	MONITORING	12.4	20.3	-7.9	31	2	46.7
belgium vs iran	MONITORING	10.2	17.7	-7.5	63	2	49.5
Brian Brobbey	MONITORING	8.8	16	-7.2	45	2	49.3